Answer Key - Chemistry - Undergraduate Level Assessment 20 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) CH 3 F; B) CH 3 Cl; C) CH 3 Br; D) CH 3 I
   Note: CH 3 Cl has a chemical shift due to the electronegativity of chlorine, which causes the hydrogen atoms in the methyl group to resonate at a frequency that is approximately 1.55 kHz downfield from TMS in a 12.0 T magnetic field.

2. **Answer:** D
   Options: A) Q = 1012; B) Q = 2012; C) Q = 3012; D) Q = 6012
   Note: The Q-factor is calculated using the formula Q = f / f, where f is the resonant frequency and f is the bandwidth; for an X-band EPR cavity, this results in a Q-factor of 6012 when the resonator bandwidth is 1.58 MHz.

3. **Answer:** D
   Options: A) Strong oxidizing conditions; B) Low-spin electron configuration; C) Metallic physical properties; D) Unpaired electrons
   Note: Unpaired electrons are crucial for both paramagnetism and ferromagnetism because they create a net magnetic moment that allows materials to respond to external magnetic fields, with unpaired electrons aligning in the same direction to produce stronger magnetic properties in ferromagnetic materials.

4. **Answer:** A
   Options: A) 17 O; B) 19 F; C) 29 Si; D) 31 P
   Note: The correct answer is 17 O because its nuclear magnetic resonance frequency of 115.5 MHz at a 20.0 T magnetic field is consistent with its gyromagnetic ratio, which determines the frequency of resonance for a given magnetic field strength.

5. **Answer:** A
   Options: A) Neutrons; B) Alpha particles; C) Beta particles; D) X-rays
   Note: Cobalt-60 is produced when cobalt-59 absorbs a neutron, resulting in a nuclear reaction that transforms it into cobalt-60 through the addition of one neutron to its nucleus.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: T_1 relaxation is not sensitive to molecular motions at the Larmor frequency; instead, it is primarily influenced by interactions between spins and their surrounding environment, while T_2 relaxation is more directly related to molecular motions.

7. **Answer:** True
   Options: A) True; B) False
   Note: This statement is true because, according to the second law of thermodynamics, the total entropy of an isolated system always increases during spontaneous processes, reflecting the natural tendency for systems to progress toward thermodynamic equilibrium.

8. **Answer:** False
   Options: A) True; B) False
   Note: Planck's quantum theory actually describes the emission spectra of diatomic molecules as consisting of discrete energy levels rather than a continuous energy distribution.

9. **Answer:** False
   Options: A) True; B) False
   Note: The hydrogen atom in CHCl 3 undergoes rapid intermolecular exchange due to hydrogen bonding and dynamic molecular interactions, leading to a single peak in its 1 H NMR spectrum rather than multiple peaks.

10. **Answer:** False
    Options: A) True; B) False
    Note: The limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule is actually 5/2 R at low temperatures due to translational and rotational degrees of freedom, and approaches 7/2 R at high temperatures when vibrational modes are also considered.

Long Form Response

11. **Answer:** The thermal stability of hydrides in group-14 elements is influenced by factors such as atomic size, bond strength, and the presence of lone pairs on the central atom. As we move from carbon (CH 4) to lead (PbH 4), the stability decreases, following the order PbH 4 < SnH 4 < GeH 4 < SiH 4 < CH 4. This trend can be attributed to the increasing size and decreasing electronegativity of the central atom, which weakens the M-H bond. For instance, CH 4 is highly stable due to the strong covalent bonds formed between carbon and hydrogen, while PbH 4 is less stable because the larger lead atom forms weaker bonds, making it more prone to decomposition. Additionally, the presence of lone pairs on the central atom can exacerbate instability in heavier hydrides.
    Note: The answer correctly identifies that the thermal stability of group-14 hydrides decreases from CH 4 to PbH 4 due to increasing atomic size and decreasing bond strength, which weaken the M-H bonds, and provides a clear stability trend supported by the characteristics of each compound.

12. **Answer:** The absorption of electromagnetic radiation by proteins at a wavelength of 190 nm is significant as it falls within the ultraviolet (UV) region of the spectrum,

where many biological molecules, including proteins, exhibit characteristic absorption properties. This wavelength is particularly relevant because it corresponds to the absorption of peptide bonds and certain side chains of amino acids, which can be crucial for understanding protein structure and function. The ability to absorb UV light at this wavelength allows for the quantification of protein concentration through techniques such as UV-Vis spectroscopy. Moreover, this absorption can indicate the presence of protein denaturation or alterations in conformation, making it a valuable tool in biochemical studies. Overall, understanding the absorption at 190 nm enhances our knowledge of protein behavior and interactions in various biological contexts.
Note: correct because it identifies the specific wavelength of 190 nm as significant for protein absorption in the UV spectrum, linking this to the absorption characteristics of peptide bonds and amino acid side chains, which are essential for analyzing protein structure and function.

End of Answer Key